

ECEN 142 Communications and Networking
MIDTERM
Open Book and Notes

Please sign the following statement upon completing this examination:

"In recognition of and in the spirit of the Honor Code, I certify that I have neither received nor given unpermitted aid on this examination."

Signature Dylan Thornburg

Print Name Dylan Thornburg

Problem 1 30 [30]

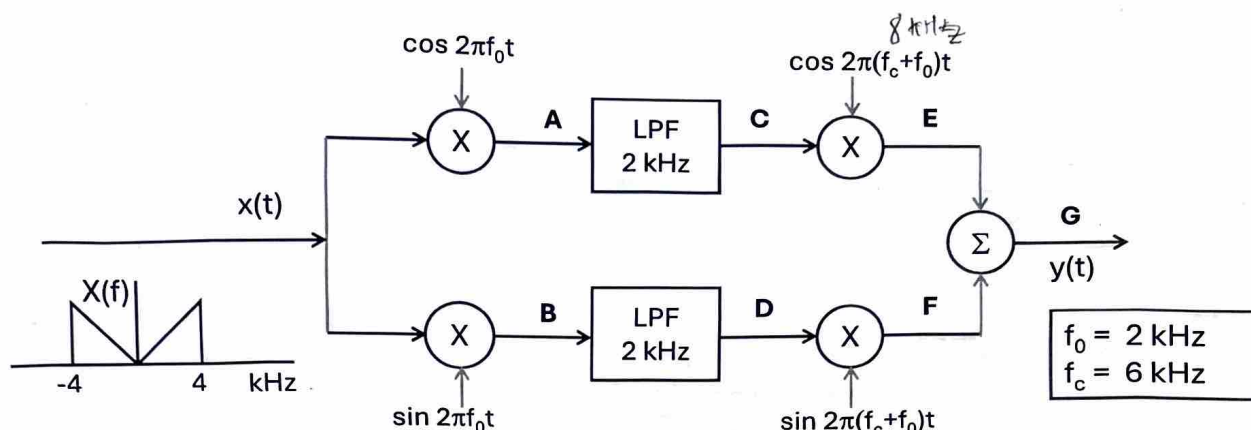
Problem 2 30 [30]

Problem 3 40 [40]

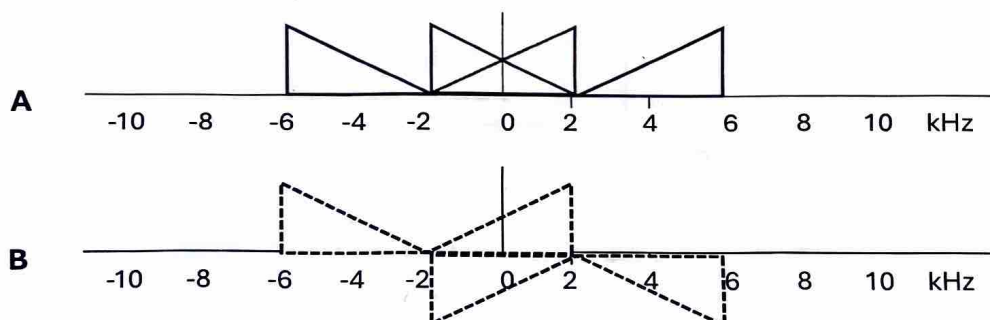
Total 100 100] Good

Problem 1 [30 points]

Consider the following system designed to modulate an analog voice signal with a nominal bandwidth of 4 kHz:

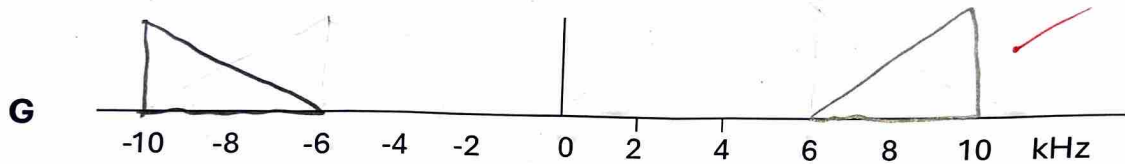
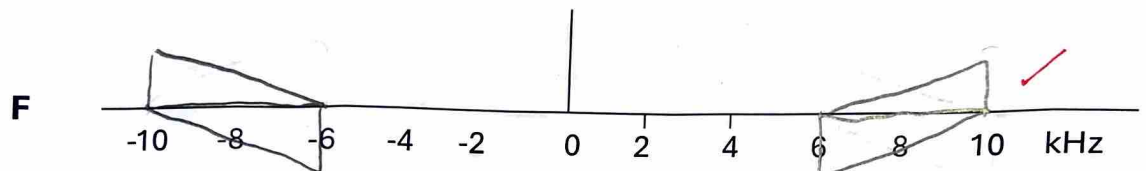
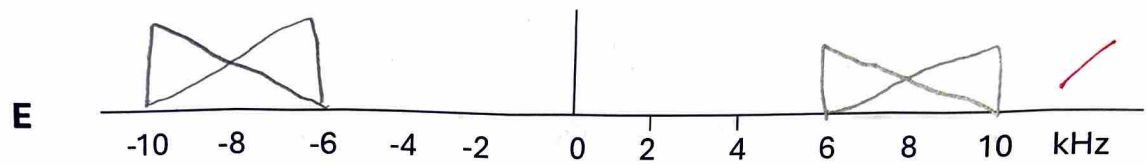
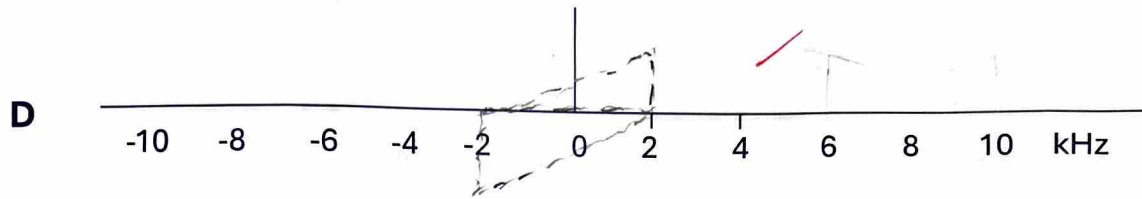
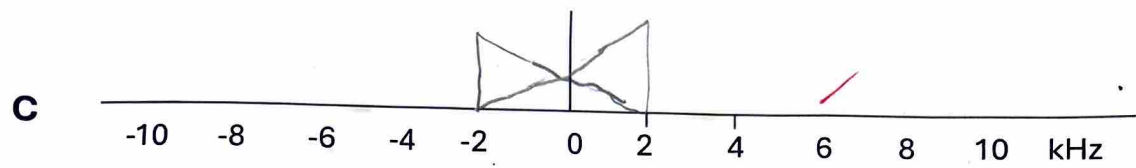


Your task is to determine the modulation type by sketching the signal spectra at various labeled points in the system diagram. In your sketches, use **solid lines** to represent the real components of the spectrum and **dotted lines** to represent the imaginary components. You may ignore gain factors. As a reference, the spectra at points A and B are shown below.



- Draw the spectra at points C, D, E, F, and G on the following page.
- Identify the type of analog modulation implemented by this circuit.

Transmits only upper sideband, so this is SSB modulation.



Problem 2 [30 points]

Digital Sum Variation (DSV) is an important parameter of line codes because it predicts the amount of baseline wander (i.e., *local DC shift*) caused by AC coupling in a transmission channel. For this reason, many line coding techniques are designed to minimize this quantity. If $a_n, n = 0, 1, 2, 3, \dots$, represents the **transmitted** symbol sequence, the **Running Digital Sum (RDS)** at time instant n is defined as

$$RDS(n) = \sum_{k=0}^n a_k$$

The Digital Sum Variation (DSV) is then defined as

$$DSV = RDS_{\max} - RDS_{\min}$$

where

$$RDS_{\max} = \max [RDS(n)] \text{ and } RDS_{\min} = \min [RDS(n)].$$

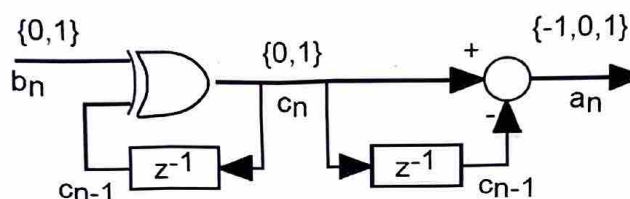
It has a longer temporary DC shift though.

(a) For the bipolar line coding shown below, assuming that the I.C. $c_{-1} = 0$, prove that

$$RDS(n) = c_n$$

(b) Find the RDS bounds and the corresponding DSV value.

(c) Evaluate RDS bounds and the DSV for the B8ZS code. Explain your results briefly.



a.

$$RDS(n) = \sum_{k=0}^n a_k, \quad a_n = c_n - c_{n-1}, \text{ so } RDS(n) = \sum_{k=0}^n (c_k - c_{k-1}) ; \text{ expanded out } \rightarrow$$

$$RDS(n) = (c_0 - c_{-1}) + (c_1 - c_0) + (c_2 - c_1) + \dots + (c_n - c_{n-1}). \text{ All will cancel except final}$$

$$c_n \text{ and } c_{-1}, \text{ so } RDS(n) = c_n - c_{-1} \text{ but } c_{-1} = 0, \text{ so } \boxed{RDS(n) = c_n} \checkmark$$

b. c_n can only be 1 or 0, so $RDS_{\max} = 1, RDS_{\min} = 0, DSV = 1 - 0 = 1$ $\checkmark$

c. B8ZS when 8-bit zero, substitute $\xrightarrow{\text{prev} + \text{pulse}} 000+ - 0 - +$, or $\xrightarrow{\text{prev} - \text{pulse}} 000- + 0 + -$

starting w/ $RDS = 1 \rightarrow 11121101$

starting w/ $RDS = 0 \rightarrow 000-10010$

$$RDS_{\max} = 2$$

$$DSV = 2 - (-1) = 3$$

$$RDS_{\min} = -1$$

4

B8ZS allows the running digital sum to move outside normal AMI ranges

briefly b/c of the violation pulses. However b/c it's balanced, it always returns to its prev AMI state afterward ($1 \rightarrow 1$), ($0 \rightarrow 0$) preventing long term DC shift.

Problem 3 [40 points]

Consider the following parity check matrix for a (7, 4) block code:

$$H = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

P^T I_3

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

(a) Construct the corresponding generator matrix G .

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{aligned} p_1 &= m_1 \oplus m_2 \oplus m_3 \\ p_2 &= m_1 \oplus m_3 \oplus m_4 \\ p_3 &= m_1 \oplus m_2 \oplus m_4 \end{aligned}$$

(b) List the sixteen code words and find the $d_{\min}$ for the code.

Input	Codeword	Input	Codeword
0000	0000 111	1000	1000 111
0001	0001 011	1001	1001 100
0010	0010 110	1010	1010 001
0011	0011 101	1011	1011 010
0100	0100 101	1100	1100 010
0101	0101 110	1101	1101 001
0110	0110 011	1110	1110 100
0111	0111 000	1111	1111 111

$$d_{\min} = \text{Hamming dist} = 3$$

(c) What are its error detection and correction capabilities?

$$d_{\min} - 1 = \text{error det.} = 2 \quad \text{correction} = \left\lfloor \frac{d_{\min} - 1}{2} \right\rfloor = 1$$

double error detection + single error correction

(d) Suppose the received codeword is $\mathbf{r} = [c_1 c_2 c_3 c_4 c_5 c_6 c_7] = [0 1 1 0 0 0 0]$. Find the syndrome $\mathbf{e}$ and the correct transmitted codeword. Explain your answer briefly. (Note that c_1, c_2, c_3, c_4 correspond to the message bits while c_5, c_6, c_7 represent the parity bits).

$[0 1 1 0 0 0 0] \rightarrow$ should be 0110 011, c_4 is the problem

$$\mathbf{e} = H\mathbf{r}^T = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ is H col 4, so c_4 (checks out) is issue!

w/ single error correction, the correct codeword is $[0 1 1 1 0 0 0]$